

IRMACS Defect Register

Every known and suspected defect, risk and improvement carried forward across the firmware sessions to date — consolidated from working memory, the changelog, the analysis archive and this session's source study.

Compiled 2026-08-14, rev 4 Release 0.6.1.14 · app 24e7bd9c7
Revert baseline pre-rules-study-20260813 · 43 repos Push state local only

3
P1 CRITICAL

13
P2 HIGH

17
P3 MEDIUM

6
P4 LOW

18
FIXED,
SESSION

43
JUNE AUDIT

P1 · Critical FLEET RISK OR DATA LOSS

Active exposure on deployed hardware, or a defect that corrupts state. These gate a release.

ID	DEFECT	COMPONENT	EVIDENCE	STATUS
RUL-01	Pushed rule can write arbitrary SRAM <code>ioset 145</code> (POKE) executes at <i>parse</i> time in <code>pcRulesParseIOSET</code> , looping <code>*Addr++ = Val</code> . Gated on <code>appPRODUCTION == 0</code> — which has never shipped above 0, so it is live in every field build.	<code>rules</code> <code>security</code>	<code>rules_parse_text.c:414</code>	• Open
OWB-01	c98c DS2482 wedged — needs a site visit Chronic transport failure. AC01 has no DS2482 reset or power GPIO, so <code>esp_restart()</code> can never clear it — proven in field (11 reboots in 12 min did not recover; a power cycle did).	<code>onewire</code> <code>hardware</code>	Field-proven, 2026-07-30	• Ops action
TBD-01	Free of a stack pointer → heap corruption <code>psEI->psUB = &sUBuf</code> (stack) then <code>free(psEI->psUB)</code> on the failure path. Verified still present. ThingsBoard path only, so it affects the KC868-A6 build, not the SiteWhere fleet.	<code>thingsboard</code>	<code>task_thingsboard.c:110,148</code>	• Open

P2 · High CORRECTNESS, OR SHIPPED BUT UNPROVEN

Wrong behaviour, missing verification, or a blocker on planned work. Several are committed but have no hardware time.

ID	ITEM	COMPONENT	EVIDENCE	STATUS
RUL-02	The parser has zero automated coverage <code>vRulesFunctionalTest()</code> is unreachable behind two closed gates — <code>cmakeDIAGS == 0</code> , and the call inside <code>vDiagsModules</code> is commented out. The parser that caused a year-long FOTA outage is untested.	<code>rules</code> <code>test</code>	<code>main.cpp:157</code> , <code>hal_diags.c:49</code>	• Open
RUL-03	Decoder hides the fields where bugs live <code>vRulesDecodeActuate</code> prints nothing for UPGRADE and REBOOT; case <code>kwIOSET</code> is commented out. This is why <code>AP0=0</code> was invisible in every <code>rules</code> dump for a year, and it rules out <code>decode→reparse</code> round-trip testing.	<code>rules</code>	<code>rules_decode.c:131</code>	• Open
RUL-04	GDI button-event rules can never fire <code>xRulesCheckEventGP_DIN()</code> is a stub: <code>{ return epRES_FALSE; }</code> . Where <code>HAL_GDI > 0</code> , every such condition evaluates false. The examples in the test corpus parse but cannot fire.	<code>rules</code>	<code>rules_engine.c:561</code>	• Open
AEP-01	Black-hole outage undetected for minutes — events lost, not buffered WiFi loss detects in seconds (<code>flagLX_STA</code>) and a dead broker via RST immediately, but silent packet loss with the link up takes <i>minutes</i> : TCP retransmits and <code>send()</code> keeps succeeding. At QoS 0 a publish "succeeds" once handed to TCP, so events are freed as sent when they were not. QoS 1 closes it — the missing PUBACK fails the publish and the drain loop re-queues to the front.	<code>sitewhere</code> <code>thingsboard</code> <code>mqtt</code>	Decision due ~2026-08-21	• Deferred
RUL-12	RULES completion gate is meaningless <code>(SenseCount + ModeCount + OtherCount + RulesCount) >= RulesSizeReq</code> sums	<code>sitewhere</code> <code>endpoints</code>	<code>sitewhere-gpb_v2.c:200</code>	• Open

`SenseCount` — a *live endpoint count*, not a GMAP counter — against a rules-requested count. At 69–104 vs a request of 3–7 it is true on the first rule received, so RULES is flagged complete before the rules arrive. `c74c` grew 69→99→104 in 22 min, so SENSE registrations double-count across cycles too.

TNT-01	FSM authentication has no hardware time Rewrite committed. Needs the full bench pass on both clients — critically the terminal-resize-during-password case, which is the scenario it exists for and which needs a NAWS-capable client.	<code>server-tnet</code>	<code>ed55a6e</code>	• Untested
CLI-01	Cursor key on an empty line kills every single-character command Root-caused. <code>cmdFlag.cli</code> is set unconditionally when a CSI/SS3 introducer arrives (<code>commands.c:387</code>) but never cleared when the sequence yields an empty line, so <code>vCommandInterpret</code> routes every keystroke into the buffer. Reached by UP with no history or DOWN past the newest entry . Not a delay — the CLI goes unresponsive until Enter. One-line minimal fix.	<code>commands</code>	<code>analysis/cli-input-defects.md</code>	• Blocks r
CLI-03	Redraw erases a command's last output line <code>xReport("\r\e[0K")</code> at <code>commands.c:427</code> runs unconditionally, including after CR. If a command's output does not end in a newline, the carriage return goes back to column 0 of that line and the erase wipes it. Likely the same defect as TNT-03.	<code>commands</code>	<code>analysis/cli-input-defects.md</code>	• Open
TST-01	Rules parse test is unreachable AND unsafe to enable Three gates (<code>cmakeDIAGS 0</code> stops <code>hal_diags.c</code> compiling at all, the call is commented out, <code>vDiagsModules</code> is all-or-nothing). Worse, it <i>executes</i> rather	<code>rules</code> <code>hal_esp32</code>	<code>analysis/rules-parse-test-gating.md</code>	• Open

than parses —
`xRulesProcessText` inserts
into the live rules table,
reconfigures sensing, perturbs the
GMAP counters and sets
`flagAPP_RULES`.

BLD-01	v507 migration is blocked — PLTOPT 1 unbuildable <code>f04_3x1m0_sw1.sdk</code> does not exist, so the one build that migrates an old-geometry mote cannot be produced. Must exist before any v507 campaign.	<code>build</code> <code>configs</code>	<code>analysis/pltopt-comparison.md</code>	• Blocker
BLD-02	Production build has never been run <code>appPRODUCTION = 1</code> compiles and links 44,880 B smaller but has never been flashed. <code>flagSTACK</code> silently removes ~12 KB of task stack across 17 tasks in a configuration never exercised. Soak-test under CLI load, do not just boot it.	<code>build</code> <code>hal_esp32</code>	<code>analysis/cli-command-audit.md</code> §3-4	• Untested
OWB-02	DS2482 log-storm fix never deployed Three root causes found for the 20.8 lines/sec flood (~190 MB/day). Fix P1 applied but left uncommitted and undeployed; needs a version bump and a baseline capture.	<code>onewire</code>	<code>c764</code> , 2026-07-30	• Stalled
TNT-02	Option negotiation can loop <code>vTelnNetNegotiate</code> answers every received option unconditionally and never consults stored state; the SGA branch replies DO to a WONT. RFC 1143 requires no response to a request that does not change the status quo. Now fixable — <code>2ed1e6a</code> made the stored state trustworthy.	<code>server-tnet</code>	<code>server-tnet.c:215</code> , RFC 1143	• Open

P3 • Medium **SURPRISING SEMANTICS, DECISIONS, DEFERRED WORK**

Real but bounded. Several are pending decisions rather than defects, and are listed so they do not get lost.

ID	ITEM	COMPONENT	EVIDENCE	STATUS
RUL-05	IOSET executes during parsing and erases itself	<code>rules</code>	<code>rules_parse_text.c:334,443</code>	• By design?

The `ioS_*` range reconfigures WiFi/MQTT, tests credentials or reads memory at parse time, then `RULE_CLR_FLAG(kwIOSET)` removes the action. So `then ioset 143 ...` fires on every rules push, not when scheduled.

RUL-06	Keywords silently accept numeric codes <code>pcRulesPeekToken</code> falls back to parsing the token as a number in the same range, so <code>sch 74</code> $\equiv$ <code>sch ip4b</code> . Undocumented; a stray number in a hand-written rule is accepted.	rules	rules_parse_text.c:88	• Open	
RUL-07	Flag-bit ceiling is inconsistent <code>pcRulesParseToken</code> uses <code>kw < 64</code> , <code>vRulesFlagSET</code> uses <code>kwNum < 63</code> , for the same bitmap. Harmless today; a trap when keywords are added.	rules	:113 vs rules_engine.c:89	• Open	
RUL-08	Slot latch is consumed by the schedule, not the action If the schedule fires and the condition then fails, the hour is spent. A mote booting inside its uptime-guard window loses the slot entirely.	rules	rules_engine.c:919	• By design?	
RUL-09	No rule editing exists <code>xRulesRuleRecall</code> , <code>Update</code> and <code>Delete</code> are all <code>return erFAILURE</code> stubs.	rules	rules_parse_text.c:944-948	• By design	
TXH-01	Publish queue holds 90 KB of mostly-empty payload <code>psUBufCreate</code> must allocate the worst case (300 B) but the <i>whole</i> 300 is queued until drained, while telemetry is 57 B. At the full 300-entry queue that is 90,000 B where ~17,000 B is needed, plus 19,200 B of constant-topic mallocs. Trimming before queueing recovers ~73 KB of outage buffering.	sitewhere	buffers	analysis/aep-tx-payload-sizing.md	• Designed
SDK-01	ESP-IDF @ -files corruption recurs Un-normalised <code>CMAKE_BINARY_DIR</code> defeats the substring anti-recursion guard when the build dir is a	esp-idf	build	espressif/esp-idf#18971	• Upstream

	symlink. Reported upstream; a local patch is carried and recorded in the release manifest.				
BLD-03	idf.port returns to the repo file Removed on the theory that <code>idf.saveScope</code> defaults to Global; the extension wrote it straight back into workspace settings. Either pin <code>saveScope</code> in user settings or accept permanent churn.	build	observed 2026-08-13	• Open	
PRT-01	printfx is in FLASH, not IRAM <code>syslog</code> carries <code>IRAM_ATTR</code> but every callee is cached flash, so the annotation cannot do what it exists for. Recorded twice; never actioned.	printfx syslog	uart-console-io-flow.md §50.2, §71	• Open	
FLS-01	MD5 policy undecided Choose: never auto-add, or gate at runtime on the bootloader description. <code>appADD_MD5 = 0</code> is load-bearing — a pre-v3.1 bootloader cannot parse an MD5 row, so this is a brick risk.	hal_esp32	hal_checks.h:37	• Decision	
MBS-01	Modbus write support — quadrant decision Read path revived, fixed and committed but never runtime-tested. Write support designed, not built, awaiting the bounded / state-managed choice (0 / 1 / 3).	modbus endpoints	analysis/modbus-codebase-study.md	• Decision	
MBS-02	Modbus per-host unit-id (S3) Outstanding item from the same workflow.	modbus	analysis/modbus-codebase-study.md	• Open	
OWB-03	Site remediation per AN148 Check DQ and GND are on the same twisted pair before any cable, reader or flyback-diode replacement — the cheapest test first.	onewire hardware	analysis/an148-applied-ac01.md	• Ops action	
OWB-04	Channel-attribution instrumentation idle <code>ds248xSTAT_DEBUG</code> counters shipped to identify which DS2482 channel wedges the device; no fleet data gathered yet.	onewire	analysis/ds2482-channel-attribution.md	• Awaiting da	

OPS-01	ioaset 67 2 1 outstanding on the fleet Coredump upload routes via <code>ioHostCONF</code> , which defaulted to a host with no HTTPS path. Deployed units need the option set or their dumps never arrive.	<code>fleet-ops</code>	v0.6.1.5 note	• Ops action
OPT-01	New-option adopt-and-persist path untested in field Options added since the last mass reflash rely on a grow-and-persist path exercised only on the bench.	<code>options</code>	v0.6.1.9 note	• Untested
TNT-03	Telnet last line garbled on CR Long-standing cosmetic defect, never started. May now be resolved by the CR/LF rework in <code>abbb9ab</code> — needs re-checking rather than assuming.	<code>server-tnet</code>	0.6.1.11 note	• Recheck

P4 · Low **CLEANUP AND DOCUMENTATION**

ID	ITEM	COMPONENT	EVIDENCE	STATUS
XCT-01	Dead NULL checks after allocation <code>CONFIG_HEAP_ABORT_WHEN_ALLOCATION_FAILS=y</code> means <code>malloc</code> can never return NULL, so ~11 existing checks are unreachable and ~34 "unchecked" sites are not bugs. Keep the config; do not add guards.	<code>cross-cutting</code>	<code>heap_caps.c</code> , 13 abort sites	• Noted
RUL-10	Reference document has eight untraced areas MODE/SENSE per-URI grammar, <code>pcEpConfigCMD</code> , <code>xID_Validate</code> , actuator field units, the annunciation vocabulary. Marked $\triangle$ and not to be relied on.	<code>rules</code> <code>docs</code>	<code>rules-module-reference.md</code> §17	• Open
RUL-11	Non-thread-safe static in <code>pcRulesFieldName()</code> Verified still present. ThingsBoard build only.	<code>rules</code>	<code>rules_engine.c:72</code>	• Open
CMN-01	TST_STDIN_TERM is dead Defined in <code>stdioX.h</code> , referenced nowhere.	<code>common</code>	<code>stdioX.h:15-23</code>	• Open
CLI-02	termSTDIN_TERM keeps the sdkconfig coupling Same class as the line-ending defect fixed in <code>abbb9ab</code> , still present in the command-buffer path. Works today, so left alone.	<code>commands</code>	<code>commands.c:762</code>	• Open
DOC-01	Release manifest does not pin the SDK <code>manifest.txt</code> records every component HEAD; the <code>esp-idf</code> line and its local patch were added by hand for 0.6.1.14 and should become part of the procedure.	<code>releases</code>	<code>releases/v0.6.1.14/</code>	• Open

Fixed this session **COMMITTED, NOT PUSHED**

Retained because several remain unverified on hardware, and because the FOTA chain took four independent fixes to restore.

COMMIT	FIX	COMPONENT		VERIFIED
5d453b4	Config RETAINED across an MQTT drop Every disconnect wiped all 171 identity records and the rules table, so the mote went blind and could grant no access until GMAP completed — 3 s at best, 60 s+ through the GMAP loop. Split so only a deliberate re-registration discards config. Verified on hardware: I=171/171/270 and an UPGRADE rule firing while disconnected.	sitewhere	thingsboard	Field ✓
1b63321	Producer never blocks on a full TX queue portMAX_DELAY stalled the rules/actuator task — the one reading iButtons — for the rest of an outage once the queue filled. Now drops and counts. Verified: Queue FULL at 300, record DROPPED with the producer continuing.	sitewhere	thingsboard	Bench ✓
b8875b4	Registration deadlock broken REGISTER published through the queue it exists to drain, so a mote could never re-register once that queue filled — the exact state a long outage produces.	sitewhere		Not yet
3af1b14	Constant topic stored, not copied Saved 20,400 B and 300 malloc/free pairs at full depth. Verified: Blk 298→601 — one allocation per event, not two.	sitewhere		Bench ✓
6b3f0d8	myASSERT tests the expression, not a truncated copy A regression introduced the same day: casting moved the test onto (i32_t)x, firing spuriously for 64-bit/float expressions. Aborted inside the V chain with output suppressed — black console, dead CLI, telnet refusing. Single-evaluation win retained (−1,360 B).	hal_esp32		Bench ✓
59876ff	GMAP watchdog counts stall, not elapsed A fixed 60 s budget tore down sessions still receiving. ID=170/170/171 is one record short; teardown discarded all 170 and re-requested all 171, failing identically every cycle — the self-perpetuating loop behind the historic syndrome. 56 timeouts, 8 motes.	sitewhere		Not yet
f7b65f2	RX buffer 300→512; no dispatch into a closed session 336 B readbuf could not receive a 253 B ruleCreate behind a 41 B header — 6 bytes of margin, 18 logged kills. And isconnected was tested outside the mutex, so RX drained a live socket into a wiped handler table. cycle_error raised to SL_ERR.	irmacos	mqtt	Not yet
47008fe	Action parameters parsed when the keyword ends the rule An early return skipped default assignment, so THEN UPGRADE passed an empty request mask — no rule-driven FOTA check had ever worked. THEN REBOOT also switched the boot partition.	rules		Field ✓
05e6a03	Second μS→Sec conversion removed	rules		Field ✓

Uptime guards needed ~9.5 years to be true.

029dcbd

Per-rule IP4B/IP14I latch; boot hour forfeited only when past the slot

rules

Field ✓

Shared static blocked a second rule; unconditional arming delayed first check by up to 95 min.

59b41e5

Conditional trace no longer fabricates values

rules

Field ✓

A union passed by value to a variadic function printed `hi32(Lo)|hi32(Val)`.

6455026

App option handler actually called

options

Bench ✓

`#ifdef` on a function name is never true, so every option exception handler was dead.

881d718

Telnet auth tests the right flag

server-tnet

Bench ✓

Gated on per-connection state that was never assigned, so the prompt never appeared.

49bcc40

Auth timeout; unconditional drop on failure

server-tnet

Bench ✓

An unbounded EAGAIN loop let one silent client lock telnet out until reboot.

2ed1e6a

Option mask no longer wipes its neighbours

server-tnet

Bench ✓

A missing `~` zeroed the other three options in the byte; ECHO and SGA share byte 0.

ed55a6e

Authentication moved into the FSM

server-tnet

Not yet

IAC mid-credential was consumed as data and echoed back as our own command.

abbb9ab

CR, LF or either pair accepted as end of line

common

Bench ✓

Terminated only on the sdkconfig UART setting, so a telnet client sending CR NUL never completed a line.

36fb021

History scans bounded; out-of-bounds read fixed

buffers

Not yet

Linear scans walked past the allocation; an unbounded copy targeted a 128-byte buffer; UP on a fresh boot echoed uninitialised heap.

A prior sweep catalogued 6 critical, 8 high, 13 medium, 6 low and 10 improvements in `analysis/BUG_ANALYSIS.md`. It predates most of the work since, so items are not carried into the counts above except where spot-checked.

REF	ITEM	COMPONENT	SPOT-CHECK
C-1 / C-2	Stack pointer freed / passed via queue	thingsboard	• Still present
C-6	Inverted assertion in <code>halFlashErasePartition()</code>	hal_esp32	• Fixed since June
H-6	Static buffer in <code>pcRulesFieldName()</code>	rules	• Still present
C-3,4,5 · H-1...8 · M-1...13 · L-1...6 · I-1...10	Remaining 40 items Not re-verified. Several are likely closed by the I2C, syslog, heap and coredump work since June.	various	• Unverified

Recommended next step for this backlog: re-verify the 40 unchecked items against current source before planning against them. Roughly a third touch files that have been substantially reworked since June, so the true open count is probably well below 43.

Sources · working memory (29 memories) · `changelog.md` · `analysis/` (28 documents) · direct source study of `z-comp/rules` (3,046 lines)

Priorities are an engineering judgement, not a customer commitment. Items marked "By design?" need a decision before they can be closed.

Revert baseline `pre-rules-study-20260813` across 43 repositories · all work committed locally, nothing pushed.